

The logic of arguments - Set 2 - Solutions

Specification Arg1-Arg4.

Answer key

Question	Answer	Topic
1	D	The logic of arguments
2	B	The logic of arguments
3	B	The logic of arguments
4	E	The logic of arguments
5	A	The logic of arguments
6	B	The logic of arguments
7	A	The logic of arguments
8	A	The logic of arguments
9	G	The logic of arguments
10	B	The logic of arguments
11	E	The logic of arguments
12	H	The logic of arguments

Worked solutions

Question 1 LGC-21 · The logic of arguments · spec Arg3

Consider the statement S : **for all real x , $x^2 - 6x + 10 > 0$.** Which of the following is true?

Answer: D

Completing the square shows the expression is at least 1 for every x .

Full method - how and why

A universal claim about a quadratic. Completing the square settles it, and the question also tests whether the **reason** given is sound.

1 Complete the square: $x^2 - 6x + 10 = (x - 3)^2 - 9 + 10 = (x - 3)^2 + 1$.

why: Halve the coefficient of x to get -3 , then correct by $-(-3)^2 = -9$.

2 $(x - 3)^2 \geq 0$, so the expression is $\geq 1 > 0$ for every real x . **True.**

why: A square is never negative; the minimum value is 1 at $x = 3$.

3 Check the false reasons: $x = 3$ gives $9 - 18 + 10 = 1$, not 0; and the discriminant $36 - 40 = -4 < 0$, so there are **no** real roots.

why: The claim that it has two real roots is simply wrong, and would in any case not prove positivity.

Where it goes wrong

- Assuming any quadratic takes negative values - one with a positive leading coefficient and negative discriminant never does.
- Mis-completing the square as $(x - 3)^2 + 10$ or $(x - 6)^2 + 1$.

Takeaway Completing the square proves positivity outright: $a(x - h)^2 + k$ with $a > 0, k > 0$ is always positive.

Question 2 LGC-22 · The logic of arguments · spec Arg2

For real numbers a, b : P : ' $ab = 0$ '; Q : ' $a = 0$ '. Which one of the following is true?

Answer: B

$a = 0 \Rightarrow ab = 0$, so P is necessary for Q . But $ab = 0$ could be because $b = 0$ with $a \neq 0$, so P is not sufficient.

Full method - how and why

This one reverses the usual direction - the check must be done carefully rather than by pattern-matching.

1 Test ' P sufficient for Q ', i.e. $ab = 0 \Rightarrow a = 0$: take $a = 5, b = 0$. Then $ab = 0$ but $a \neq 0$. **Not sufficient.**

why: The zero-product rule gives $a = 0$ or $b = 0$, not $a = 0$.

2 Test ' P necessary for Q ', i.e. $Q \Rightarrow P$: if $a = 0$ then $ab = 0 \cdot b = 0$. **True, so necessary.**

why: Anything times zero is zero.

3 So P is necessary but not sufficient for Q .

why: Note the direction: the **weaker** condition ($ab = 0$) is necessary for the stronger one ($a = 0$).

Where it goes wrong

- Answering 'sufficient but not necessary' by reflex - in this question the roles are swapped from the common pattern.
- Thinking $ab = 0$ requires both to be zero.

Takeaway $ab = 0 \Rightarrow a = 0$ or $b = 0$; and always check both directions explicitly rather than assuming the familiar answer.

Question 3 LGC-23 · The logic of arguments · spec Arg1

Consider the statement: ****if a and b are irrational, then $a + b$ is irrational.**** Which of the following pairs is a counterexample?

Answer: B

A counterexample needs the hypothesis true (both irrational) and the conclusion false ($a + b$ rational):

$$\sqrt{2} + (-\sqrt{2}) = 0.$$

Full method - how and why

A counterexample must make the hypothesis **true** and the conclusion **false**. Check both halves for each candidate.

1 Hypothesis: both a and b irrational. Conclusion to break: $a + b$ irrational - so we need $a + b$ **rational**.

why: Only a case satisfying the hypothesis can refute the statement.

2 $a = \sqrt{2}$, $b = -\sqrt{2}$: both irrational ✓, and $a + b = 0$, which is rational ✓. **Counterexample**.

why: $-\sqrt{2}$ is irrational: if it were rational, so would its negative be.

3 Reject $a = 1$, $b = -1$: these are rational, so the hypothesis fails and nothing is refuted.

why: A case that does not satisfy the hypothesis says nothing about the implication.

4 Reject $\sqrt{2} + \sqrt{3}$, $\sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$ and $\pi + \sqrt{2}$: each sum is irrational, so the conclusion holds and the statement is not broken.

why: These are examples **of** the statement, not against it.

Where it goes wrong

- Picking rational numbers whose sum is rational - the hypothesis is not satisfied.
- Picking irrationals whose sum is irrational - the conclusion is not broken.

Takeaway A counterexample satisfies the 'if' and violates the 'then'. Check both halves every time.

Question 4 LGC-24 · The logic of arguments · spec Arg4

Let S be the statement: ****every even number greater than 2 can be written as the sum of two primes.**** A single counterexample to S would be

Answer: E

To refute 'for all n , $P(n)$ ' you exhibit one n with $P(n)$ false.

Full method - how and why

What it takes to refute a universal statement - and what it takes to refute ***this*** one in particular.

1 The statement has the form 'for all even $n > 2$, $P(n)$ '.

why: Identify the quantifier and the domain: even numbers greater than 2.

2 Its negation is 'there exists an even $n > 2$ with $P(n)$ false'.

why: One failure suffices; you do not have to show the property fails often.

3 So a counterexample is a single even number greater than 2 that cannot be written as a sum of two primes.

why: It must lie in the stated domain **and** fail the property.

4 (This is Goldbach's conjecture - no counterexample is known, which is why it remains a conjecture.)

why: Verifying billions of cases is not a proof; one counterexample would be a disproof.

Where it goes wrong

- Offering an **odd** number that is not a sum of two primes - outside the stated domain, so irrelevant.
- Offering an even number that **is** a sum of two primes - that confirms the statement, it does not refute it.
- Thinking many confirming examples add up to a proof.

Takeaway To disprove 'for all', produce one object **in the domain** for which the property fails; confirming instances never prove it.

Question 5 X2-24 · The logic of arguments · spec Arg2

For a real number x , P : ' $x > 0$ '. Q : ' $x^3 > 0$ '. Which one of the following is true?

Answer: A

Cubing preserves sign, so $x > 0 \Leftrightarrow x^3 > 0$.

Full method - how and why

Cubing preserves sign in both directions, so this is an equivalence - unlike squaring.

1 If $x > 0$ then $x^3 = x \cdot x \cdot x > 0$: **sufficient**.

why: A product of three positives is positive.

2 Conversely, if $x \leq 0$ then $x^3 \leq 0$ (negative cubed is negative, and $0^3 = 0$). So $x^3 > 0$ forces $x > 0$: **necessary**.

why: Proved by contrapositive - check the negative and zero cases rather than hunting for a counterexample.

3 So P is necessary and sufficient.

why: Cubing is a strictly increasing, sign-preserving operation.

Where it goes wrong

- Importing the squaring counterexample ($x = -2$) - for cubes $(-2)^3 = -8 < 0$, so it does not apply.
- Answering 'sufficient but not necessary' by pattern-matching to x^2 questions.

Takeaway Odd powers preserve sign and order; even powers do not. $x > 0 \Leftrightarrow x^3 > 0$.

Question 6 LGC-05 · The logic of arguments · spec Arg3

Which of the following statements is/are true?

I For every real number x , there exists a real number y such that $y > x$.

II There exists a real number y such that for every real number x , $y > x$.

III For every real number x , there exists a real number y such that $xy = 1$.

Answer: B

I is true (take $y = x + 1$). II claims a single largest real number exists: false. III fails for $x = 0$.

Full method - how and why

Quantifier **order** matters. ' $\forall x \exists y$ ' and ' $\exists y \forall x$ ' say completely different things.

1 I: for each x , can I find a bigger y ? Take $y = x + 1$. **True**.

why: y is allowed to depend on x , because x is chosen first.

2 II: is there a single y bigger than **every** real x ? That would be a largest real number. Taking $x = y$ contradicts $y > x$. **False**.

why: Here y is fixed **before** x , so it must work for all x at once - including $x = y$.

3 III: for each x is there y with $xy = 1$? For $x = 0$, $0 \cdot y = 0 \neq 1$. **False**.

why: 'For every x ' includes the awkward cases; zero has no reciprocal.

4 So I only.

why: Match to the list.

Where it goes wrong

- Reading I and II as the same statement - the difference is the order of the quantifiers, nothing else.
- Forgetting $x = 0$ in III and accepting $y = \frac{1}{x}$ uncritically.

Takeaway In $\forall x \exists y$ the y may depend on x ; in $\exists y \forall x$ one y must serve every x . And 'for every' always includes the edge cases.

Question 7 LGC-06 · The logic of arguments · spec Arg1

A teacher says: 'You will pass only if you revise.' Which of the following statements means the same thing?

Answer: A

'A only if B' means ' $A \Rightarrow B$ ': passing implies revising. It does not promise that revising guarantees a pass.

Full method - how and why

The phrase 'only if' is the one piece of logical English almost everyone mistranslates. 'A only if B' means $A \Rightarrow B$.

- 1 Let A = 'you pass', B = 'you revise'. The sentence is 'A only if B'.

why: Label first.

- 2 'A only if B' says that A can happen **only** in the case that B holds - so A forces B : $A \Rightarrow B$.

why: Compare with 'A if B', which is the other direction $B \Rightarrow A$. 'Only if' and 'if' point opposite ways.

- 3 Translate back: **if you pass, then you revised.**

why: Revising is necessary for passing, but the teacher promises no guarantee that it is enough.

Where it goes wrong

- Reading it as 'if you revise, you will pass' - the converse, and a much stronger promise than the teacher made.
- Reading it as 'if and only if'; 'only if' gives one direction only.
- Choosing 'if you do not pass then you did not revise' - the inverse, not equivalent (the correct contrapositive is 'if you did not revise, you did not pass').

Takeaway 'A if B' means $B \Rightarrow A$; 'A only if B' means $A \Rightarrow B$. 'Only if' marks the **necessary** condition.

Question 8 LGC-12 · The logic of arguments · spec Arg3

Consider the statement: ****there exists a real number k such that for all real x , $x^2 + kx + 1 > 0$.**** Which of the following is true?

Answer: A

Need discriminant $k^2 - 4 < 0$, i.e. $-2 < k < 2$. So some (not all) values of k work; $k = 0$ gives $x^2 + 1 > 0$. ($k = 2$ gives $(x + 1)^2$, which is 0 at $x = -1$.)

Full method - how and why

An $\exists \forall$ statement: you only need to produce **one** k that works for **all** x . Then check why the offered reasons are right or wrong.

- 1 Try $k = 0$: $x^2 + 1 > 0$ for every real x , since $x^2 \geq 0$.

why: One successful witness proves an existential statement.

- 2 So the statement is true. Now check the reason offered with each claim.

why: In a 'true because...' question both the verdict **and** the justification must be right.

- 3 Is it true for **every** k ? No: the condition is discriminant < 0 , i.e. $k^2 - 4 < 0$, so only $-2 < k < 2$.

why: A positive quadratic with no real roots never crosses the axis.

- 4 And $k = 2$ gives $(x + 1)^2$, which equals 0 at $x = -1$ - so that witness fails the strict inequality.

why: The boundary case of the discriminant gives ≥ 0 , not > 0 .

- 5 So: true, with $k = 0$ as the witness.

why: Match to the list.

Where it goes wrong

- Claiming it holds for all k - $k = 5$ gives $x^2 + 5x + 1$, which is negative at $x = -1$.
- Offering $k = 2$ as the witness, which fails only at the single point $x = -1$ - easy to miss.
- Thinking a quadratic with a positive x^2 coefficient must eventually be negative; it is large **positive** for large $|x|$.

Takeaway To prove $\exists$, exhibit one witness - but check the witness satisfies the condition **strictly** where required.

Question 9 LGC-16 · The logic of arguments · spec Arg3

Which of the following statements is/are true about real numbers?

- I For all x and all y , if $x < y$ then $x^2 < y^2$.
- II For all x there exists y such that $x < y$ and $x^2 < y^2$.
- III There exist x and y with $x < y$ and $x^2 > y^2$.

Answer: G

I fails for $x = -2$, $y = 1$. II: take $y = |x| + 1$. III: $x = -2$, $y = 1$ works.

Full method - how and why

Three quantified statements about squaring and ordering. Negative numbers decide all three.

- 1 I: $x < y \Rightarrow x^2 < y^2$ for all reals? Take $x = -2$, $y = 1$: $-2 < 1$ but $4 > 1$. **False.**
why: Squaring is only order-preserving for non-negative numbers.
- 2 II: for each x , is there y with $x < y$ and $x^2 < y^2$? Take $y = |x| + 1$: then $y > x$ and $y > |x|$ so $y^2 > x^2$. **True.**
why: Choosing y larger than $|x|$ guarantees both conditions at once - and y may depend on x .
- 3 III: do some $x < y$ have $x^2 > y^2$? The counterexample from I works: $x = -2$, $y = 1$. **True.**
why: Reuse the work - the same pair settles both.
- 4 So II and III.
why: Match to the list.

Where it goes wrong

- Accepting I by testing only positive numbers.
- Rejecting II by picking a bad y ; you get to choose y , so choose a helpful one.

Takeaway $x < y \Rightarrow x^2 < y^2$ needs $0 \leq x$; and to prove $\forall x \exists y$, construct y as a formula in x .

Question 10 LGC-18 · The logic of arguments · spec Arg4

A statement reads: ****for every positive integer n , there is a prime p with $n < p < 2n$.**** Which of the following is its negation?

Answer: B

Negate the quantifiers in turn: 'for every n , there exists p with ...' becomes 'there exists n such that for every prime p , not ...', i.e. no prime lies strictly between n and $2n$.

Full method - how and why

Negating a nested $\forall \exists$ statement: work from the outside in, flipping one quantifier at a time.

- 1 Formalise: $\forall n \exists p$ (prime) with $n < p < 2n$.
why: Two quantifiers, so two flips.
- 2 Flip the outer: $\exists n$ such that $\neg[\exists p$ with $n < p < 2n]$.
why: 'For all' fails exactly when one case fails.
- 3 Flip the inner: $\exists n$ such that $\forall$ primes p , $\neg(n < p < 2n)$ - i.e. **no** prime lies strictly between n and $2n$.
why: 'There exists' fails exactly when every candidate fails.
- 4 So: there is a positive integer n for which no prime p satisfies $n < p < 2n$.
why: (The original is Bertrand's postulate and is in fact true, so this negation is false - but the question is about form.)

Where it goes wrong

- Keeping 'for every n ' and only negating the inside - that is a much stronger claim, not the negation.
- Turning it into ' $\exists n$ and $\exists p$ with $p \leq n$ or $p \geq 2n$ ', which is trivially true and says nothing: the inner quantifier must become **for all** primes.

Takeaway Negate nested quantifiers outside-in: each $\forall$ becomes $\exists$, each $\exists$ becomes $\forall$, and the innermost statement is negated once.

Question 11 LGC-20 · The logic of arguments · spec Arg1

Which of the following statements is/are logically equivalent to 'if it is raining, then the ground is wet'?

- I It is not raining, or the ground is wet.
- II If the ground is not wet, then it is not raining.
- III If the ground is wet, then it is raining.

Answer: E

' $A \Rightarrow B$ ' is equivalent to 'not A , or B ' (I) and to its contrapositive (II). III is the converse, which is not equivalent.

Full method - how and why

Two standard equivalents of $A \Rightarrow B$: the contrapositive, and the disjunctive form 'not A , or B '.

- 1 I: 'not A or B '. An implication only fails when A is true and B false - exactly when ' $\neg A$ or B ' fails. **Equivalent.**
why: Check the one bad case: raining and ground dry makes both statements false; every other case makes both true.
- 2 II: 'if the ground is not wet, it is not raining' - the contrapositive. **Equivalent.**
why: Swap and negate.
- 3 III: 'if the ground is wet, it is raining' - the converse. **Not equivalent;** the ground could be wet from a burst pipe.
why: A concrete alternative cause is the cleanest refutation.
- 4 So I and II.
why: Match to the list.

Where it goes wrong

- Rejecting I because it looks nothing like an implication - test it against the single case that breaks an implication.
- Accepting III because wet ground is evidence of rain; evidence is not implication.

Takeaway $A \Rightarrow B$ is equivalent both to 'not A , or B ' and to $\neg B \Rightarrow \neg A$. The converse is never among the equivalents.

Question 12 LGC-09 · The logic of arguments · spec Arg1

Consider the statement S : **for all real x , if $x > 0$ then $x + \frac{1}{x} \geq 2$.** Which of the following statements is/are true?

- I S is true.
- II The converse of S (for all real x , if $x + \frac{1}{x} \geq 2$ then $x > 0$) is true.
- III The contrapositive of S is true.

Answer: H

For $x > 0$, $x + \frac{1}{x} - 2 = \frac{(x-1)^2}{x} \geq 0$, so S is true, and so is its contrapositive (III). For the converse: if $x < 0$ then $x + \frac{1}{x} < 0 < 2$, so $x + \frac{1}{x} \geq 2$ forces $x > 0$ (II true).

Full method - how and why

One inequality, examined three ways. The contrapositive is free once the statement is settled; the converse must be tested separately.

- 1 I: for $x > 0$, write $x + \frac{1}{x} - 2 = \frac{x^2 - 2x + 1}{x} = \frac{(x-1)^2}{x}$.
why: Put over a common denominator x - the numerator turns out to be a perfect square.
- 2 Since $(x-1)^2 \geq 0$ and $x > 0$, the fraction is ≥ 0 , so $x + \frac{1}{x} \geq 2$. **True.**
why: A non-negative numerator over a positive denominator is non-negative. Equality at $x = 1$.
- 3 III: the contrapositive of a true statement is automatically **true**.
why: No work needed - contrapositive and original are logically equivalent.
- 4 II (the converse): suppose $x + \frac{1}{x} \geq 2$; must $x > 0$? If $x < 0$ then both x and $\frac{1}{x}$ are negative, so $x + \frac{1}{x} < 0 < 2$ - the hypothesis fails. And $x = 0$ is not allowed. So $x > 0$. **True.**
why: Proved by eliminating the alternatives, which is itself a contrapositive argument.
- 5 So all three.
why: Match to the list.

Where it goes wrong

- Assuming the converse is false because converses 'usually' are - here it genuinely holds.
- Multiplying the inequality $x + \frac{1}{x} \geq 2$ by x without knowing the sign of x ; that is precisely what is in question.

Takeaway $x + \frac{1}{x} \geq 2$ for $x > 0$ is worth knowing; and a statement always shares its truth value with its contrapositive, never necessarily with its converse.